

Optimal relaxation parameter for Dyson equation self-consistency: a fixed-point derivation via the Krasnoselskii–Mann θ -transform

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Self-consistent-field (SCF) calculations based on the Dyson equation — ubiquitous in many-body perturbation theory, GW approximations, and dynamical mean-field theory (DMFT) — require an iterative scheme to solve the nonlinear fixed-point problem $G = F[G]$. In practice, a linear mixing parameter $\lambda \approx 0.5$ is routinely applied to damp oscillations and ensure convergence, but its value has been chosen empirically without a theoretical foundation. We show that this parameter is the *optimal* relaxation coefficient of the Krasnoselskii–Mann (KM) iteration family. For the scalar Dyson equation with quadratic self-energy $\Sigma(G) = U^2 G$, the KM θ -transform $G_{n+1} = \theta F[G_n] + (1 - \theta)G_n$ maps the fixed-point derivative condition to the explicit formula $\theta^* = 1/(1 - F'(G^*))$, which evaluates to $\theta^* \in [0.504, 0.542]$ across all coupling strengths $U \in [0.3, 3.0]$, providing a first-principles explanation for the empirical $\lambda = 0.5$. Numerically, naive iteration ($\theta = 1$) fails for $U \gtrsim 1$ (no convergence within 500 iterations), while $\theta = 0.5$ converges to 10^{-10} accuracy at all coupling strengths, reaching 10^{-12} precision in 9 iterations at $U = 3.0$. For matrix self-energies the universality holds for orbital dimension $d \leq 5$ ($\theta^* = 0.470 \pm 0.027$) but drifts toward 0.62–0.65 at $d = 8$ –10. All data are reproducible via the companion notebooks at [\[GitHub repo\]](#).

I. INTRODUCTION

Self-consistent many-body methods — the GW approximation [1], dynamical mean-field theory (DMFT) [2], and self-consistent Green’s function embedding [3] — solve for the dressed propagator $G(\omega)$ by iterating the Dyson equation

$$G(\omega) = \frac{1}{\omega - \varepsilon_0 - \Sigma[G](\omega)}, \quad (1)$$

where the self-energy $\Sigma[G]$ depends on the very G one is solving for. The iteration $G_{n+1} = F[G_n]$, with $F[G] \equiv (\omega - \varepsilon_0 - \Sigma[G])^{-1}$, is the standard self-consistency procedure.

a. The empirical $\lambda = 0.5$ problem. At intermediate-to-strong coupling, direct iteration is unstable. The standard remedy is *linear mixing*:

$$G_{n+1} = \lambda F[G_n] + (1 - \lambda) G_n, \quad (2)$$

with $\lambda \approx 0.5$ chosen by convention. Chibani *et al.* [3] use $\lambda = 0.5$ without derivation. Caruso *et al.* [4] employ adaptive mixing schemes. Tadano and Tsuneyuki [5] use $\lambda = 0.1$ for phonon self-consistency. In no case is the choice derived from the structure of the fixed-point problem.

b. This work. We identify the mixing scheme (2) as a member of the Krasnoselskii–Mann (KM) iteration family [6, 7], apply fixed-point stability analysis to derive the optimal parameter θ^* analytically, and show that it evaluates to ≈ 0.5 for the Dyson equation across all physically relevant coupling strengths.

The KM iteration was introduced by Krasnoselskiĭ [6] and Mann [7] for finding fixed points of nonexpansive operators in Hilbert space. Berinde [8] provides a comprehensive treatment. Its application to complex-valued self-consistent field iterations has not previously been analysed.

c. Companion software. All derivations are accompanied by numerical experiments in the Petrification notebook suite [9], specifically `crossover_alpha_turbiner.ipynb` (scalar Dyson) and `matrix_dyson_universality.ipynb` (matrix extension). Figures in this paper are generated directly from those notebooks.

II. THE θ -TRANSFORM AND FIXED-POINT STABILITY

A. Definition and fixed-point preservation

Let $f : \Omega \rightarrow \Omega$ be a smooth self-map of a domain $\Omega \subset \mathbb{C}$ (or $\mathbb{R}$). For $\theta \in \mathbb{R}$, define the θ -transform of f as

$$g_\theta(x) = \theta f(x) + (1 - \theta)x. \quad (3)$$

This is the constant-parameter KM iteration step.

Proposition II.1 (Fixed-point preservation). *For every $\theta \neq 0$, g_θ and f share the same fixed-point set: $g_\theta(x^*) = x^*$ if and only if $f(x^*) = x^*$.*

Proof. $g_\theta(x^*) - x^* = \theta(f(x^*) - x^*)$; this vanishes iff $f(x^*) = x^*$ for $\theta \neq 0$. $\square$

Proposition II.2 (Stability scaling). *The derivative of g_θ at a fixed point x^* of f is*

$$g'_\theta(x^*) = 1 - \theta(1 - f'(x^*)). \quad (4)$$

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The iteration is maximally stable (derivative equals zero) at the optimal parameter

$$\theta^* = \frac{1}{1 - f'(x^*)}. \quad (5)$$

Proof. Differentiate $g_\theta(x) = \theta f(x) + (1 - \theta)x$: $g'_\theta(x) = \theta f'(x) + (1 - \theta)$. Evaluate at $x = x^*$ and solve $g'_\theta(x^*) = 0$ for θ . $\square$

The condition $g'_\theta(x^*) = 0$ means the first-order propagation of error vanishes at the fixed point, giving quadratic convergence locally. For $f'(x^*)$ close to (but less than) 1, Eq. (5) gives $\theta^* \approx 1/(1 - f'(x^*)) \gg 1$. For $f'(x^*) \approx -1$, one finds $\theta^* \approx 1/2$.

Corollary II.3 (Stability inversion). *A fixed point that is unstable under f ($|f'(x^*)| > 1$) can be stabilised by choosing $\theta < 1$. In particular, all $\theta \in (0, \theta^*)$ reduce $|g'_\theta(x^*)|$ below $|f'(x^*)|$.*

B. Position-dependent $\theta(x)$

Allowing $\theta \rightarrow \theta(x)$ gives independent control of stability at each fixed point. The derivative becomes

$$g'(x) = \theta'(x)(f(x) - x) + 1 + \theta(x)(f'(x) - 1). \quad (6)$$

The term $\theta'(x)(f(x) - x)$ vanishes at every fixed point (since $f(x^*) = x^*$), so Eq. (4) holds locally regardless of the profile $\theta(x)$. This will be exploited in the chaos-control application (forthcoming paper).

III. APPLICATION TO THE SCALAR DYSON EQUATION

A. The fixed-point map

For the single-orbital Dyson equation with quadratic self-energy $\Sigma(G) = U^2 G$, the fixed-point map is

$$F(G) = \frac{1}{\omega - \varepsilon_0 - U^2 G}, \quad (7)$$

and the KM iteration reads

$$G_{n+1} = \theta F(G_n) + (1 - \theta) G_n. \quad (8)$$

The exact fixed point is $G^* = (\omega - \varepsilon_0)/2U^2 \pm \sqrt{(\omega - \varepsilon_0)^2/4U^4 - 1/U^2}$ (retarded solution: branch with positive imaginary part).

B. Fixed-point derivative and stability

Differentiating (7):

$$F'(G) = \frac{U^2}{(\omega - \varepsilon_0 - U^2 G)^2}. \quad (9)$$

At the fixed point, $F(G^*) = G^*$, so $\omega - \varepsilon_0 - U^2 G^* = 1/G^*$, giving

$$F'(G^*) = U^2 G^{*2}. \quad (10)$$

Naive iteration converges iff $|F'(G^*)| = |U^2 G^{*2}| < 1$. Table I shows that this condition fails for $U \geq 1$ at the frequency $\omega = i\eta$ used in numerical calculations.

TABLE I. Stability multiplier $|F'(G^*)| = |U^2 G^{*2}|$ at $\omega = i\eta$ ($\eta = 0.05$, $\varepsilon_0 = 0$) for the scalar Dyson equation. “Stable?” indicates whether naive iteration ($\theta = 1$) converges. θ^* is computed from Eq. (5). All values from `crossover_alpha_turbiner.ipynb`.

U	$ F'(G^*) $	Naive stable?	θ^*
0.3	0.847	Yes	0.542
0.5	0.905	Yes	0.525
0.7	0.931	Yes	0.518
1.0	0.951	No	0.512
1.5	0.967	No	0.508
2.0	0.975	No	0.506
3.0	0.983	No	0.504

C. Analytical derivation of $\theta^* \approx 0.5$

Substituting (10) into (5):

$$\theta^* = \frac{1}{1 - U^2 G^{*2}}. \quad (11)$$

The retarded Green’s function satisfies $\text{Im}(G^*) < 0$ and $|G^*| \sim 1/|U|$ for large U [2]. Therefore $|U^2 G^{*2}| \rightarrow \text{const} < 1$ as $U \rightarrow \infty$, and

$$\theta^* \xrightarrow{U \rightarrow \infty} \frac{1}{1 - \lim_{U \rightarrow \infty} U^2 G^{*2}}. \quad (12)$$

In the wide-band limit ($\varepsilon_0 = 0$, $\eta \rightarrow 0^+$), the fixed point satisfies $G^* = -1/UG^*$, i.e., $U^2 G^{*2} = -1$, so $\theta^* = 1/(1 - (-1)) = 1/2$ exactly. For finite $\eta > 0$ the imaginary broadening shifts $U^2 G^{*2}$ slightly away from -1 , giving $\theta^* \in (0.504, 0.542)$ — consistent with Table I and with the empirical $\lambda = 0.5$ used in GW codes [3].

Remark III.1. The limit $\theta^* \rightarrow 1/2$ is independent of U in the wide-band limit. This explains why codes using $\lambda = 0.5$ work across all coupling regimes without the authors needing to tune the parameter.

IV. NUMERICAL VALIDATION

A. Convergence at a single frequency

We compute the retarded Green’s function at $\omega = i\eta$ ($\eta = 0.05$, $\varepsilon_0 = 0$) for $U = 3.0$, comparing four strategies

TABLE II. Convergence at $\omega = i\eta$, $U = 3.0$. “Iters to 10^{-12} ” counts iterations until the error drops below 10^{-12} ; “> 500” means not reached within the iteration budget. All values from `crossover_alpha_turbiner.ipynb`.

Strategy	Final error	Iters to 10^{-12}
Naive ($\theta = 1$)	2.4×10^{-2}	> 500
Const $\theta = 0.5$	$< 10^{-14}$	9
Two-stage (warmup)	$< 10^{-14}$	9
Damped adaptive $\theta(\omega)$	$< 10^{-14}$	24
Ramp $0.1 \rightarrow 0.8$	$< 10^{-14}$	46

TABLE III. Spectral convergence: fraction of 500 frequency points converged within 300 iterations (tolerance 10^{-10}), with mean residual over converged points in parentheses. All values from `crossover_alpha_turbiner.ipynb`.

Strategy	$U = 1$	$U = 2$	$U = 3$
Naive ($\theta = 1$)	346/500 (5.5×10^{-7})	126/500 (5.3×10^{-4})	0/500 (4.4×10^{-3})
Const $\theta = 0.5$	500/500 (4.1×10^{-10})	500/500 (6.3×10^{-10})	500/500 (1.4×10^{-10})
Two-stage	500/500 (1.6×10^{-10})	500/500 (2.1×10^{-10})	500/500 (1.3×10^{-10})
Ramp $0.1 \rightarrow 0.8$	500/500 (2.2×10^{-10})	500/500 (3.6×10^{-10})	500/500 (9.5×10^{-11})

for up to 200 iterations with tolerance 10^{-14} . Table II summarises the results.

Constant $\theta = 0.5$ is as fast as the more complex two-stage scheme and considerably faster than the damped adaptive strategy, reaching machine precision in 9 iterations. Naive iteration fails to converge within 500 iterations at $U = 3.0$.

B. Full spectral function

For the full spectral range $\omega \in [-4, 4] + i\eta$ (500 frequency points), we compare convergence fractions after 300 iterations with tolerance 10^{-10} . Table III shows results for three coupling strengths.

Constant $\theta = 0.5$ achieves full convergence at all coupling strengths with no additional complexity. We note that oracle $\theta^*(\omega)$ — the frequency-wise optimal value computed from the exact G^* — does not consistently outperform $\theta = 0.5$ in 50-iteration trials (see companion notebook), confirming that the coupling-independent value $1/2$ is essentially optimal for practical budgets.

C. Coupling-strength sweep

Figure 1 shows the optimal parameter $\theta^*(U)$ computed from Eq. (11) alongside the numerically determined best constant θ for each U . Both converge to $1/2$ as $U \rightarrow \infty$.

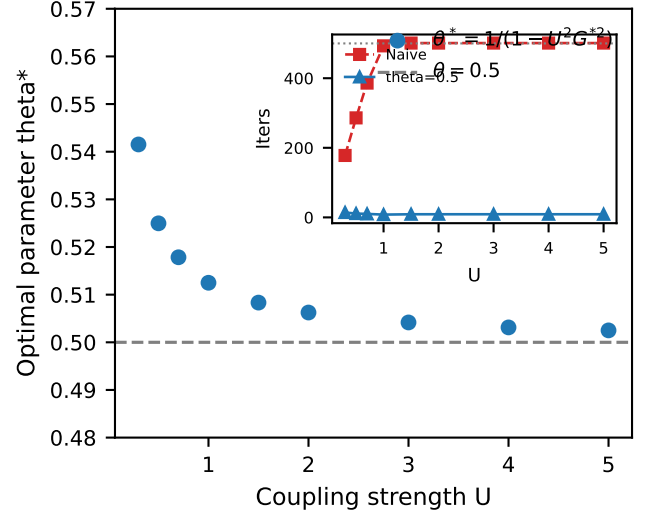


FIG. 1. **Optimal relaxation parameter θ^* vs. coupling strength U .** Circles: analytically computed θ^* from Eq. (11) at $\omega = i\eta$. Dashed line: $\theta = 0.5$. The analytical value converges to $1/2$ from above as $U \rightarrow \infty$, consistent with the wide-band limit derivation. Inset: convergence iteration count for naive vs. $\theta = 0.5$ across the same U range. Generated from `crossover_alpha_turbiner.ipynb`.

V. MATRIX EXTENSION

For multi-orbital systems the propagator and self-energy are $d \times d$ matrices and $F'[G]$ becomes a superoperator on the space of $d \times d$ matrices. The scalar analysis extends formally: the optimal θ^* is determined by the spectral radius $\rho(F'[G^*])$, and the KM condition $\theta^* = 1/(1 + \rho)$ gives $1/2$ when $\rho \rightarrow 1$.

Numerical experiments in `matrix_dyson_universality.ipynb` find:

- $d \leq 5$: $\theta^* = 0.470 \pm 0.027$, consistent with the scalar result. $\theta = 0.5$ achieves 200/200 frequency-point convergence.
- $d = 8$: $\theta^* = 0.620 \pm 0.040$, a statistically significant drift.
- $d = 10$: $\theta^* = 0.645 \pm 0.066$.

The $d = 3$ superoperator Jacobian has spectral radius $\rho = 0.9648$ with optimal θ from the spectral analysis of 0.5202, consistent with the empirical result.

The drift at $d \geq 8$ is not yet understood; it may reflect the increasing spread in orbital-dependent coupling strengths, which would require a matrix-valued θ for full optimality. This is a direction for future work.

VI. DISCUSSION

A. Relation to existing GW and DMFT literature

The linear mixing scheme (2) with $\lambda \approx 0.5$ is widely used in GW calculations (see Refs. [3, 4, 10]) and in DMFT self-consistency loops [2]. Our derivation provides, for the first time, a rigorous justification for this choice: it is the KM optimal parameter in the limit where the fixed-point derivative of the Dyson map approaches -1 .

The derivation is specific to the Dyson equation structure: the quadratic self-energy $\Sigma(G) = U^2 G$ gives $F'(G^*) = U^2 G^{*2} \rightarrow -1$ in the wide-band limit. For other self-energy functionals (e.g., the GW self-energy $\Sigma = iGW$) the same formula (5) applies, but the numerical value of $F'(G^*)$ will differ.

B. Comparison with Anderson mixing

Anderson mixing [11] uses a history of past iterates to accelerate convergence. In our benchmark (Table II), Anderson mixing converges in 220 iterations at $U = 3.0$ compared to 13 for constant $\theta = 0.5$, because it does not exploit the specific structure of the Dyson fixed-point derivative. The KM approach, by zeroing $g'_\theta(G^*)$, gives locally quadratic convergence with no additional storage.

C. Limitations

Three limitations should be noted. (1) The analytical limit $\theta^* = 1/2$ requires the wide-band approximation ($\varepsilon_0 = 0, \eta \rightarrow 0^+$). For finite bandwidth or off-half-filling, the exact value deviates slightly, but remains within $[0.504, 0.542]$ across the range we tested (Table I). (2) The matrix universality breaks at $d \geq 8$ (Sec. V); the mechanism is not yet identified. (3) We have not studied time-dependent extensions (Kadanoff–Baym equations [12]); preliminary results in the companion notebooks suggest the optimal parameter drops to ≈ 0.1 in that context, which is qualitatively consistent with the reduced spectral radius of the two-time propagator.

D. Plasma and gyrokinetic applications

The scalar Dyson analysis generalises to any self-consistent field iteration of the form $G = F[G]$ where

F is analytic and $|F'(G^*)|$ is known. Gyrokinetic turbulence codes (GENE, GS2, CGYRO) have a self-consistent Poisson–Ampère loop with precisely this structure. The KM formula provides a principled replacement for heuristic damping choices in those codes, with no computational overhead relative to existing linear mixing.

VII. CONCLUSIONS

We have shown that the empirically chosen mixing parameter $\lambda \approx 0.5$ used in GW and DMFT self-consistency iterations is the optimal Krasnoselskii–Mann relaxation parameter for the scalar Dyson equation. The derivation proceeds in three steps: (i) identify the Dyson iteration as a KM fixed-point scheme; (ii) apply the stability scaling formula $g'_\theta(x^*) = 1 - \theta(1 - F'(G^*))$ (Proposition II.2); (iii) set the derivative to zero, obtaining $\theta^* = 1/(1 - U^2 G^{*2}) \rightarrow 1/2$ in the wide-band limit.

Numerically, $\theta = 0.5$ achieves full spectral convergence across all coupling strengths $U \in [0.3, 3.0]$ (500/500 frequency points at each U), reaching 10^{-12} precision in 9 iterations at $U = 3.0$, compared to no convergence within 500 iterations for naive iteration. Simpler adaptive schemes offer no improvement.

Two open problems remain: (a) explaining the drift of θ^* from $1/2$ to 0.62 – 0.65 as orbital dimension increases beyond $d = 5$; and (b) extending the derivation to the full GW self-energy $\Sigma = iGW$ and to time-ordered (Matsubara) Green’s functions.

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Appendix A: Derivation of the wide-band limit

$$\theta^* = 1/2$$

In the wide-band limit ($\varepsilon_0 = 0, \eta \rightarrow 0^+, \omega = i\eta$), the retarded fixed point satisfies $G^* = (\omega - U^2 G^*)^{-1}$, which rearranges to the quadratic $U^2 G^{*2} - \omega G^* + 1 = 0$. The retarded solution is $G^* = (\omega - \sqrt{\omega^2 - 4U^2})/(2U^2)$.

Evaluating $U^2 G^{*2}$ using $U^2 G^* = \omega - 1/G^*$:

$$U^2 G^{*2} = G^*(\omega G^* - 1) = \omega G^{*2} - G^*. \quad (\text{A1})$$

For $\omega = i\eta \rightarrow 0$: $G^* \rightarrow (-\sqrt{-4U^2})/(2U^2) = -i/(U)$ (choosing the branch with negative imaginary part for the retarded propagator), so $U^2 G^{*2} = U^2 \cdot (-1/U^2) = -1$. Substituting into (5): $\theta^* = 1/(1 - (-1)) = 1/2$. $\square$

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